Standard Operating Procedure for Optical Transport Failure on KR-ACC-82 at KR Puram Access Site Running Arista Access Equipment

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2260

Description

The Optical Transport Failure on KR-ACC-82 at KR Puram Access site running Arista Access equipment is primarily caused by the degradation of SFP/XFP modules or fiber cuts leading to Rx/Tx power loss. This failure domain occurs due to the inherent nature of optical transport systems, which rely on the integrity of the optical signal transmitted over fiber optic cables. The Arista Access equipment, being a Layer 2/3 switch, is responsible for transmitting and receiving optical signals, and any degradation or loss of these signals can result in a major alarm, such as ALARM_2260. The root cause of this failure can be attributed to either hardware failure, such as SFP/XFP module degradation or fiber cuts, or environmental factors, such as temperature fluctuations or optical signal attenuation.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and that all necessary personnel are present.

- Confirm that the redundant link is functioning correctly.

- Perform a visual inspection of the SFP/XFP modules and fiber optic cables to check for any signs of physical damage or degradation.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss Due to Environmental Factors	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface tengigabitethernet 1/1 detail, Expected Output: SFP status, optical signal quality, and link status, Description: This command will provide detailed information about the SFP/XFP module, including its status, optical signal quality, and link status, which can help identify any issues with the module or the fiber optic cable.

Command: show log | include ALARM_2260, Expected Output: Alarm log with timestamp and description, Description: This command will display the alarm log with the timestamp and description of the ALARM_2260 event, which can provide valuable information about the cause of the failure.
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```
Command: show inventory, Expected Output: Device inventory with serial numbers and descriptions, Description: This command will provide a list of devices in the system, including their serial numbers and descriptions, which can help identify any hardware issues with the SFP/XFP modules or fiber optic cables.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP Module, Command: replace system internal 1/1, Description: Replace the SFP/XFP module with a new one and verify that the link is established and the signal is stable.

Step 2: Step: Repair or Replace Fiber Optic Cable, Command: no shutdown interface tengigabitethernet 1/1, Description: Repair or replace the fiber optic cable and verify that the link is established and the signal is stable.

Step 3: Step: Adjust Optical Signal Quality, Command: set interface tengigabitethernet 1/1 optical signal quality 0, Description: Adjust the optical signal quality to the recommended value and verify that the signal is stable.

Post-Check Verification

```
Command: show interface tengigabitethernet 1/1 detail, Expected Output: SFP status, optical signal quality, and link status, Description: Verify that the SFP/XFP module is functioning correctly and the link is established.

Command: show log | include ALARM_2260, Expected Output: Alarm log with timestamp and description, Description: Verify that the ALARM_2260 event has been cleared.

Command: show inventory, Expected Output: Device inventory with serial numbers and descriptions, Description: Verify that the device inventory is up-to-date and accurate.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Module, Command: replace system internal 1/1, Description: Restore the original SFP/XFP module and verify that the link is established and the signal is stable.

- Step: Restore Original Fiber Optic Cable, Command: shutdown interface tengigabitethernet 1/1, Description: Restore the original fiber optic cable and verify that the link is established and the signal is stable.

- Step: Restore Original Optical Signal Quality, Command: unset interface tengigabitethernet 1/1 optical signal quality, Description: Restore the original optical signal quality and verify that the signal is stable.